



Heart Disease Prediction

using Deep Learning

A neural network that predicts heart disease
from a patient's routine health checkup

Aleti Vivek Reddy

Deep Learning — Project-Based Learning (PBL)

THE PROBLEM

What are we trying to solve?

Heart disease is a leading cause of death.

- Doctors look at many test results to judge risk.
- That takes time and expert skill.
- Early warning can save lives.

Our question:



**Can a computer
learn to spot
heart disease?**

In simple words

We give the computer past patient records and let it learn the pattern of 'healthy' vs 'has disease' — so it can judge a new patient.

THE DATA

What the model learns from

303

patient records
(UCI Cleveland dataset)

13 in → 1 out

13 health measurements
predict 1 answer: disease? yes / no

A few of the inputs:

Age

Sex

Chest pain type

Blood pressure

Cholesterol

Max heart rate

Blood sugar

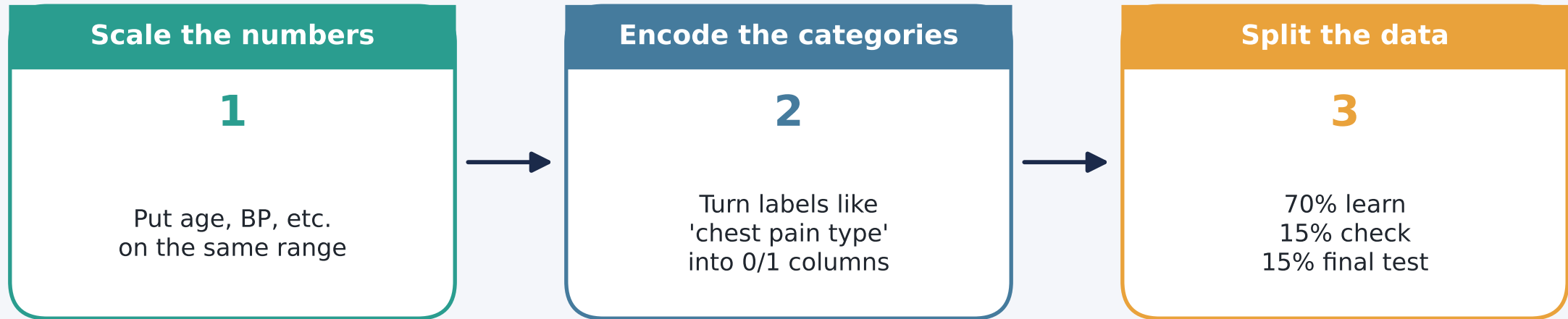
ECG result

In simple words

Think of each patient as one row in an Excel sheet: 13 numbers describe them, and the last column says whether they had heart disease.

STEP 1 — PREPARING THE DATA

Cleaning before learning

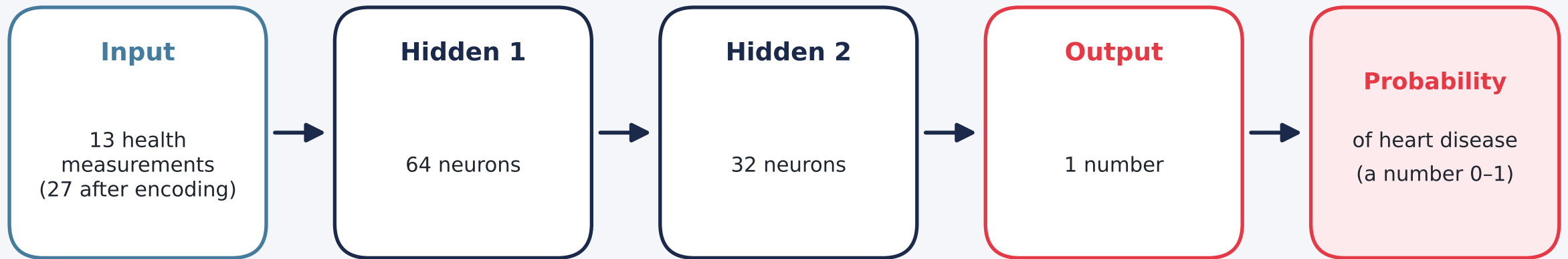


In simple words

Raw data is messy. We tidy it so every feature is comparable, then keep a part of the data hidden to fairly test the model later.

STEP 2 — THE MODEL

A simple neural network

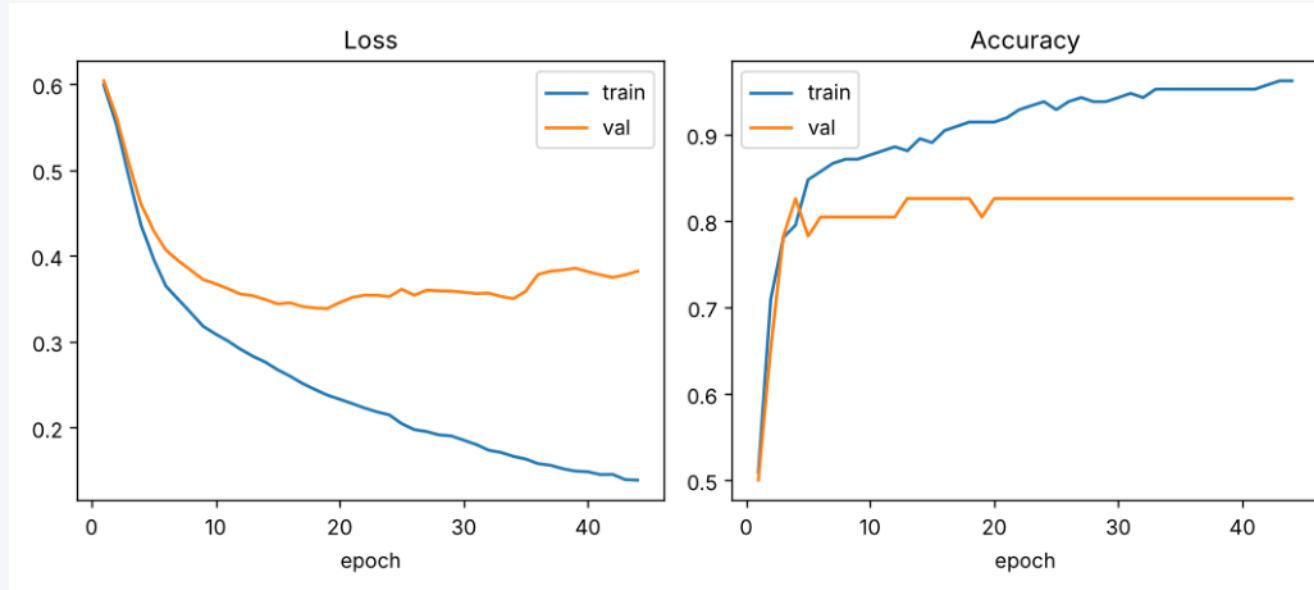


In simple words

Each layer mixes the numbers and passes them on, learning the hidden pattern. The last layer gives one score: higher = more likely to have heart disease.

STEP 3 — TRAINING

How the model learns



The idea

- Model makes a guess
- We measure the error
- It adjusts to do better
- Repeat many times

In simple words

The line going down means the model is making fewer mistakes each round. We stop automatically when it stops improving (no over-studying).

THE RESULTS

How well does it work?

82%

Accuracy

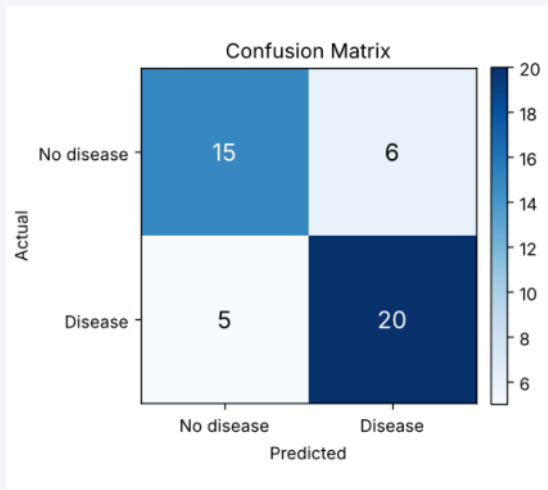
0.88

ROC-AUC

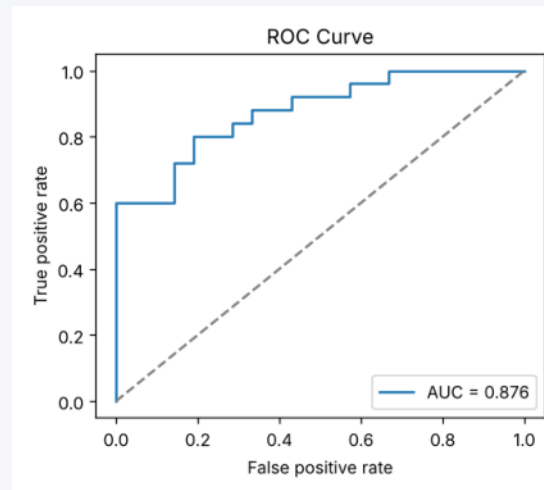
80%

Recall

(5-fold cross-validation —
a fair, stable score)



Confusion matrix



ROC curve

What it means

It catches 80% of real patients with disease.

0.88 AUC = it separates sick vs healthy well.

Confusion matrix shows correct vs wrong guesses.

CONCLUSION

What we built & learned

What we did

- Built a neural network in PyTorch
- Trained it on 303 patient records
- Tested it fairly with cross-validation
- Compared it to classic ML models

Key takeaways

- ~82% accuracy, 0.88 AUC
- Catches 80% of true cases
- On small data, simple models can match deep learning too

Next steps:

tune settings · explain predictions (SHAP) · try more data

Thank you!